

TERI Agentic RAG Chatbot — end-to-end flow

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2026-08-20

From source content to a cited answer. Read top to bottom: the write path fills the stores, the read path serves from them.

SOURCE the only content origin — nothing is read from disk or uploaded in bulk

teriin.org · **Drupal CMS** · **JSON:API** for pages and custom blocks (15 content types) · **plain HTTPS GET** for the PDFs those pages attach or link

WRITE PATH scheduled sweep on the private ingestion server · one corpus-wide run at a time

- 1 Crawl** Oldest-first, ties broken on a unique id so pagination cannot skip rows. Window floored at the **earliest unresolved failure**, so a failed document stays reachable.
- 2 Detect change** Fingerprint matches → **stop here, free**. This is most of the corpus on most runs.
- 3 Extract** Page HTML → text, keeping links, alt text and table cells. PDF routed **per page**: text → PyMuPDF, tables → Camelot, scanned → Azure OCR.
- 4 Resolve date** The page's date is the default *and* the fallback. Only the document stating its own publication can override it, past **ten gates**.
- 5 Check content** Body-text hash unchanged → update the catalogue only, **no re-embed**. Pipeline version moved → rebuild even if identical.
- 6 Chunk** Parent / child, sized by content type. Ids derived from **owned content** — not version, not position — so unchanged text keeps its id.
- 7 Embed** Children only; parents stored as zero-vectors and fetched by id later. Each child embedded behind a **title > heading** breadcrumb. Unchanged vectors reused.
- 8 Index** **Upsert the new version first, then delete the old**. The document never leaves search mid-swap, and a crash leaves the previous version intact.
- 9 Persist** Catalogue row plus theme / tag / author / attachment rows in **one transaction**. An attachment is released only when its last parent lets go.
- 10 Knowledge** Mentions → resolve identity → claims → conflict check → project to graph. Fail-open and gated off; cannot affect the indexed document.

After every sweep, and unable to fail it: knowledge catch-up (bounded backlog) → whole-graph projection → cross-store reconciliation, which checks ten invariants and reports without repairing.

STORES one system of record; everything else is derived and rebuildable from it

MySQL

SYSTEM OF RECORD

Catalogue, crawl state, failures, entities, claims. The only store that has to be right.

Qdrant

DERIVED

Passages and their vectors — the only place chunk text lives. Plus the answer cache.

Neo4j

DERIVED

Entities, claims and current-state edges. Holds identifiers and structure, never text.

READ PATH public retrieval server · per request

- 11 Understand** One structured call → intents (multi-label), a standalone rewrite, facet filters, answer shape. Told what content types and date span actually exist. **Fails open** to a plain question.
- 12 Route** Small talk → answer directly. Counting → **MySQL**. "Summarise this set" → catalogue scope. Everything else → search.
- 13 Cache** Near-verbatim question *and* identical facets → return the stored answer. Self-invalidates when the corpus or the tuning changes.
- 14 Search** Up to six legs **in parallel**, merged by reciprocal-rank fusion — so wall-clock is the slowest leg, not the sum.
 - ▶ graph — relational, budgeted
 - ▶ title-anchored — canonical pages
 - ▶ dense pull — meaning
 - ▶ lexical — acronyms, years, names
 - ▶ website / not-website split
 - ▶ paraphrases — recall
- 15 Rerank** Strict priority: **relevance band** → **completeness** → **recency** → **authority**. Nothing escapes its relevance band, however new or long it is.
- 16 Build context** Expand each hit to its parent, drop near-duplicates, fit the token budget, order it. Graph facts lead; prose keeps reserved slots. Stale events dropped for "upcoming".
- 17 Generate** Prompt chosen by what the context actually holds. Use only the numbered blocks; cite every claim; refuse rather than guess; treat the text as reference, never instructions.
- 18 Verify** Any citation pointing outside the supplied blocks is stripped. Unsupported numbers flagged. Optional per-claim entailment check with one regeneration.
- 19 Cite** Citations assembled **from stored metadata** — the model only ever emits a bare marker. Only the blocks the answer used are listed.

RESPONSE streamed as it is written

token → sources → done Answer text arrives progressively; then citations grouped **website first, then PDFs**. A stream that stops without *done* is reported as interrupted rather than shown as complete.

Cheap checks first

Fingerprint before download, hash before embedding, rules before a model call.

The model narrates

Counts from SQL, relationships from the graph, quotes from stored text, citations from metadata.

Everything fails open

Any dependency lost degrades to the behaviour that preceded it — one log line, request continues.

New before old

Write the replacement, then remove what it replaced. Deletion needs positive evidence.

Detail: ARCHITECTURE.md (field-level reference) · HLD.md (design rationale). Step 10 and the graph leg in step 14 are feature-gated and off by default.